

MAHESH REDDY CHANGAL

(667) 433 1954 | mahesh6273766@gmail.com | github.com/Soul-Eater69 | [LinkedIn](#) | [Portfolio](#)

Education

University of Maryland, Baltimore County

Master's in Computer Science

Aug 2023 – May 2025

GPA: 3.7 / 4.0

Experience

Tiger Analytics

AI Engineer

Chicago, IL

Jan 2026 – Present

- Building **production LLM systems** for enterprise users that combine **RAG**, **Azure OpenAI**, **Neo4j knowledge graphs**, and document processing pipelines (PDF, DOCX, PPT) to classify business ideas, enrich context, and identify impacted products end-to-end.
- Designed **LLM reliability patterns** including hybrid retrieval (BM25 + dense embeddings + semantic reranking), **context window management**, confidence thresholds, structured JSON outputs, and deterministic fallbacks to reduce hallucinations and ensure traceable answers in regulated enterprise settings.
- Built full-stack services with **TypeScript**, **Python FastAPI**, and **PostgreSQL**-backed metadata stores, owning the API, retrieval layer, and integration surfaces from data ingestion through generation and evaluation.
- Developed **LLM evaluation harnesses** tracking precision/recall, ranked retrieval metrics, prompt regressions, and failure modes; debugged production retrieval and generation issues under tight delivery pressure across ingestion, chunking, retrieval, and generation stages.
- Drove **tool-use and agent** patterns through prompt optimization, candidate verification, feedback-loop design, and traceable logging, exploring **MCP-style** architectures for connecting LLMs to enterprise data sources.
- Collaborated with product owners, data engineers, and offshore teams through **design reviews**, **code reviews**, and release discussions, owning problems end-to-end with focus on testability, cost awareness, and security controls.

University of Maryland, Baltimore County

Graduate Assistant & Grader

Baltimore, MD

Aug 2024 – May 2025

- Reviewed 200+ student submissions in Data Science and Network Security courses, evaluating **Python**, Java, and C++ code for correctness, algorithms, readability, edge cases, and performance.
- Mentored students on data structures, algorithms, debugging, and optimization, translating complex technical feedback into clear, actionable guidance.

Tiger Analytics

Senior Software Engineer

Hyderabad, India

Jun 2022 – Jul 2023

- Architected a no-code Terraform automation platform using **Next.js-style React**, **TypeScript**, Flask, and MongoDB, helping 500+ users visually generate cloud infrastructure workflows and reducing deployment effort by 40%.
- Built reusable full-stack components, REST APIs, and **webhook-driven** integration flows that improved maintainability across teams; supported **CI/CD**, Docker-based deployments, and Azure DevOps release workflows on cloud infrastructure.
- Developed a SQL visualization and lineage tool that mapped tables, joins, and dependencies across complex queries, improving analyst productivity by 30% and helping teams debug data relationships faster in production.
- Partnered with product managers, DevOps engineers, and UX designers to refine requirements, review technical designs, and deliver roadmap features in a fast-moving, ownership-driven engineering environment.

Tiger Analytics

Data Engineer

Hyderabad, India

Nov 2021 – May 2022

- Rebuilt and optimized distributed ETL workflows using **Spark**, PySpark, Databricks, and Airflow while supporting migration of critical data pipelines from Azure Synapse/Cosmos DB toward **Google Cloud Platform**.
- Orchestrated large-scale **event-driven** batch processing pipelines handling 5TB+ daily data, improving scheduling reliability, failure recovery, and performance through job tuning, monitoring, and workflow automation in production.
- Integrated structured and unstructured data sources through secure APIs and governed data workflows, maintaining 99.9% uptime and reducing infrastructure cost by 25%.

ICRISAT

Research Intern

Hyderabad, India

Jun 2019 – Jul 2019

- Built a machine-learning model using TensorFlow, Pandas, and NumPy to predict fertilizer application rates, improving recommendation efficiency by 15%.
- Created a Django and React platform hosted on **AWS** to serve real-time research updates to 200+ employees and support data-driven workflows.

Projects

PaperMind – Document Intelligence & RAG Platform | *Python, FastAPI, ChromaDB, Mistral-7B, LangChain, React*

- Built a full-stack **LLM-native** document intelligence platform with PDF/DOCX ingestion, chunking, embeddings, vector search, and source-grounded answers, applying **RAG patterns** for classification, summarization, and Q&A over legal- and contract-style documents.
- Implemented **LLM reliability controls** including retrieval evaluation, context-window management, citation grounding, and modular FastAPI services to reduce hallucinations and improve answer trustworthiness in production-style usage.

Code Indexer – Semantic Search & Tool-Use Backend | *Python, FAISS, Sentence-Transformers, AST Parsing, REST API*

- Created a semantic code-search engine that parses repositories with AST analysis, generates embeddings, and serves low-latency search through a **REST API** usable as an LLM tool-use backend (**MCP-style** integration surface).
- Designed reusable ingestion and chunking patterns for multi-language codebases, improving discoverability of functions, modules, and implementation details for AI agents and developers.

Conversational AI Platform – Streaming LLM Chat | *TypeScript, React, Node.js, OpenAI API, WebSockets*

- Engineered a real-time chat application with **streaming LLM responses**, conversation state, token-aware API calls, and prompt templates for reliable LLM interactions, deployed on **Vercel**-style cloud hosting.
- Implemented backend APIs and **WebSocket** flows to support responsive UX, reusable UI components, and a scalable full-stack TypeScript architecture suitable for AI-native product surfaces.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, SQL, Java, C#

AI/ML & LLMs: LLM Integration, RAG, LangChain, LangFlow, Prompt Engineering, Embeddings, Semantic Search, AI Agents, Multi-Agent Patterns, MCP (Model Context Protocol), Tool Use, Hallucination Mitigation, Context Window Management, LLM Evaluation

Full-Stack & Frontend: Next.js, React, Node.js, TypeScript, FastAPI, Flask, Django, REST APIs, GraphQL, WebSockets, Webhooks, Streaming, Event-Driven Architectures

Databases & Search: PostgreSQL, Supabase-style Postgres workflows, MongoDB, Redis, Elasticsearch, ChromaDB, FAISS, Azure AI Search, Vector Databases, Hybrid Search, Semantic Reranking

Cloud & Infrastructure: Vercel, AWS, GCP, Azure, Railway-style deployments, Docker, Kubernetes, Terraform, CI/CD, Airflow, Spark, PySpark, Databricks, Microservices

Engineering Practices: Production Debugging, System Design, Code Reviews, Technical Documentation, Agile Delivery, End-to-End Ownership, Security, Privacy, Data Governance